EEE-301
Data and Telecommunication

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Chapter 5

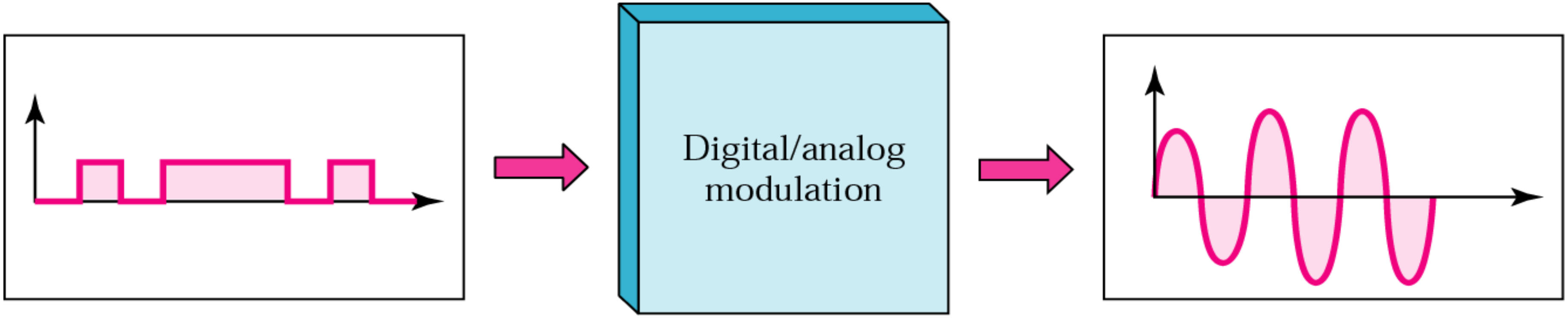
Analog Transmission

Digital to Analog Modulation

Modulation Devices

Analog to Analog Modulation

Digital to Analog Modulation



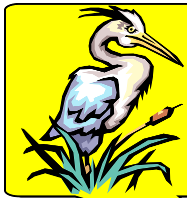
Digital-to-analog modulation: The process of changing one of the characteristics of an analog signal based on the information in a digital signal.

- A sine wave is defined by three characteristics: amplitude, frequency, and phase.
- When we vary anyone of these characteristics, we create a different version of that wave.

Digital to Analog Modulation

- So, by changing one characteristic of a simple electric signal, we can use it to represent digital data.
- Any of the three characteristics can be altered in this way, giving us at least three mechanisms for modulating digital data into an analog signal: amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK).

Bit Rate vs. Baud Rate



Note:

***Bit rate* is the number of bits per second**

- More important in speaking of computer efficiency

***Baud rate* is the number of *signal units* per second that are required to represent those bits**

-More important in speaking of data transmission

-Determine the bandwidth required to send the signal

Analogy in transportation: a baud is analogous to a car while a bit is analogous to a passenger (1: male, 0: female). The number of cars determines the traffic; that of passengers does not

Baud Rate Example

An analog signal carries 4 bits in each signal unit. If 1000 signal units are sent per second, find the baud rate and the bit rate

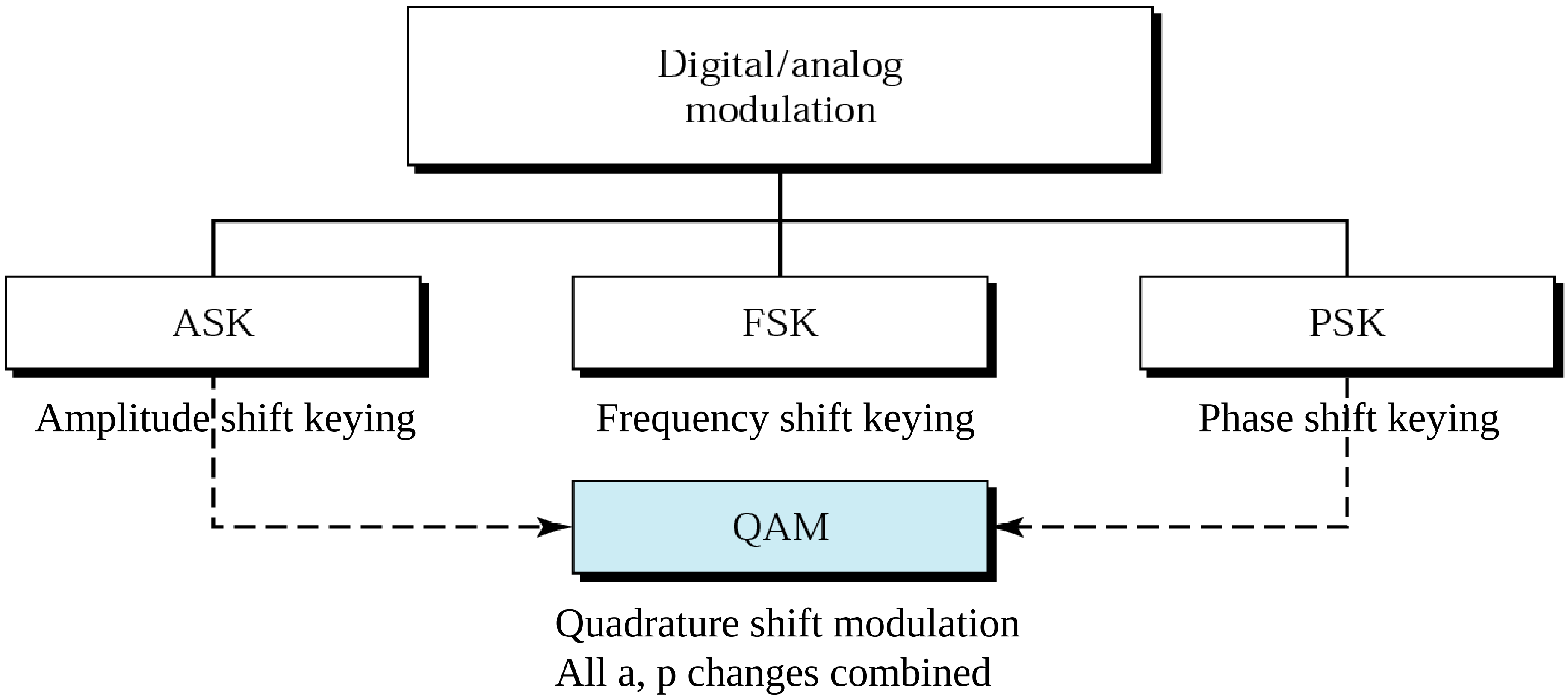
Baud rate = 1000 bauds per second (baud/s)

Bit rate = 1000 x 4 = 4000 bps

The bit rate of a signal is 3000. If each signal unit carries 6 bits, what is the baud rate?

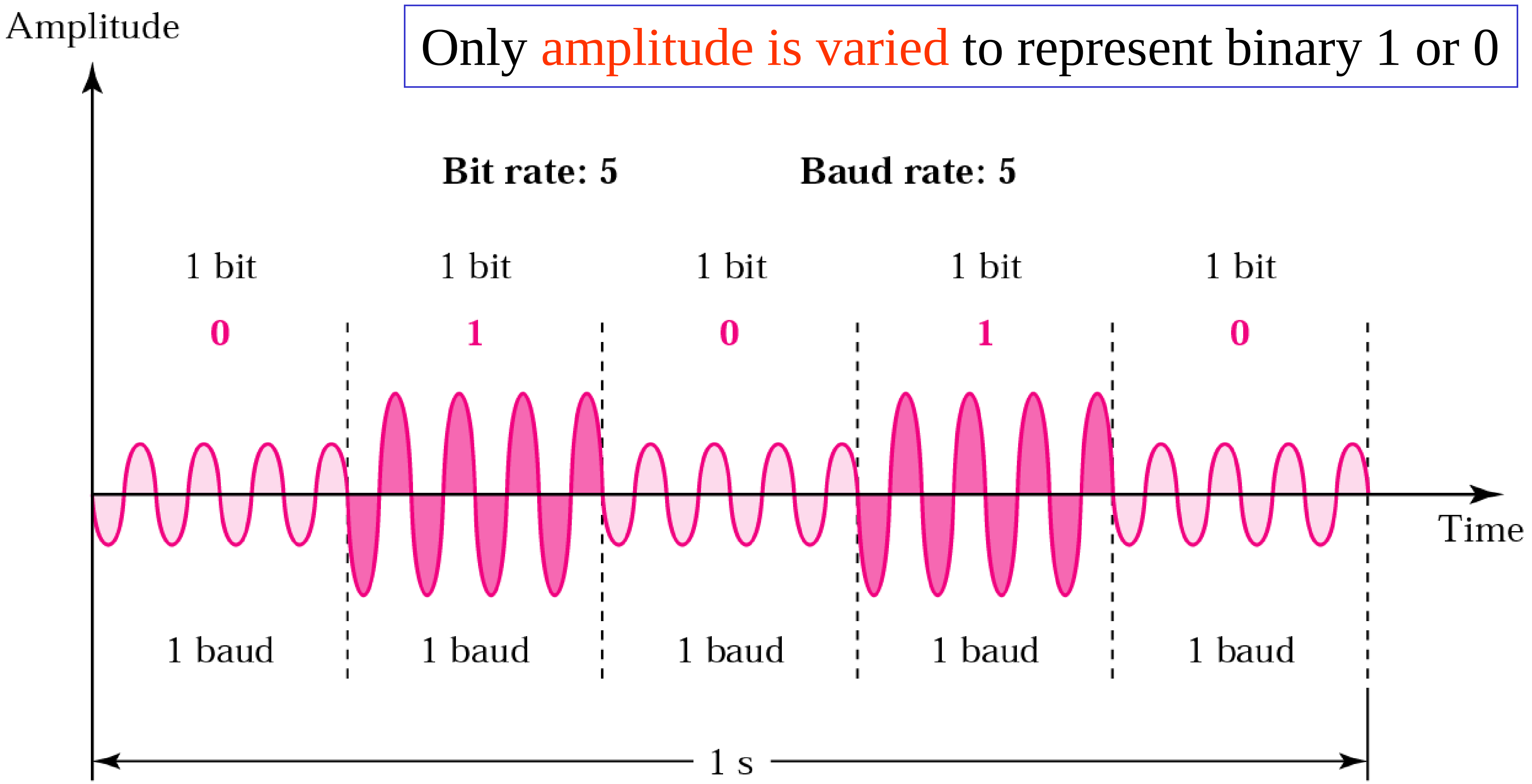
Baud rate = 3000 / 6 = 500 baud/s

Digital-Analog Modulation Schemes



Amplitude Shift Keying (ASK)

Only amplitude is varied to represent binary 1 or 0



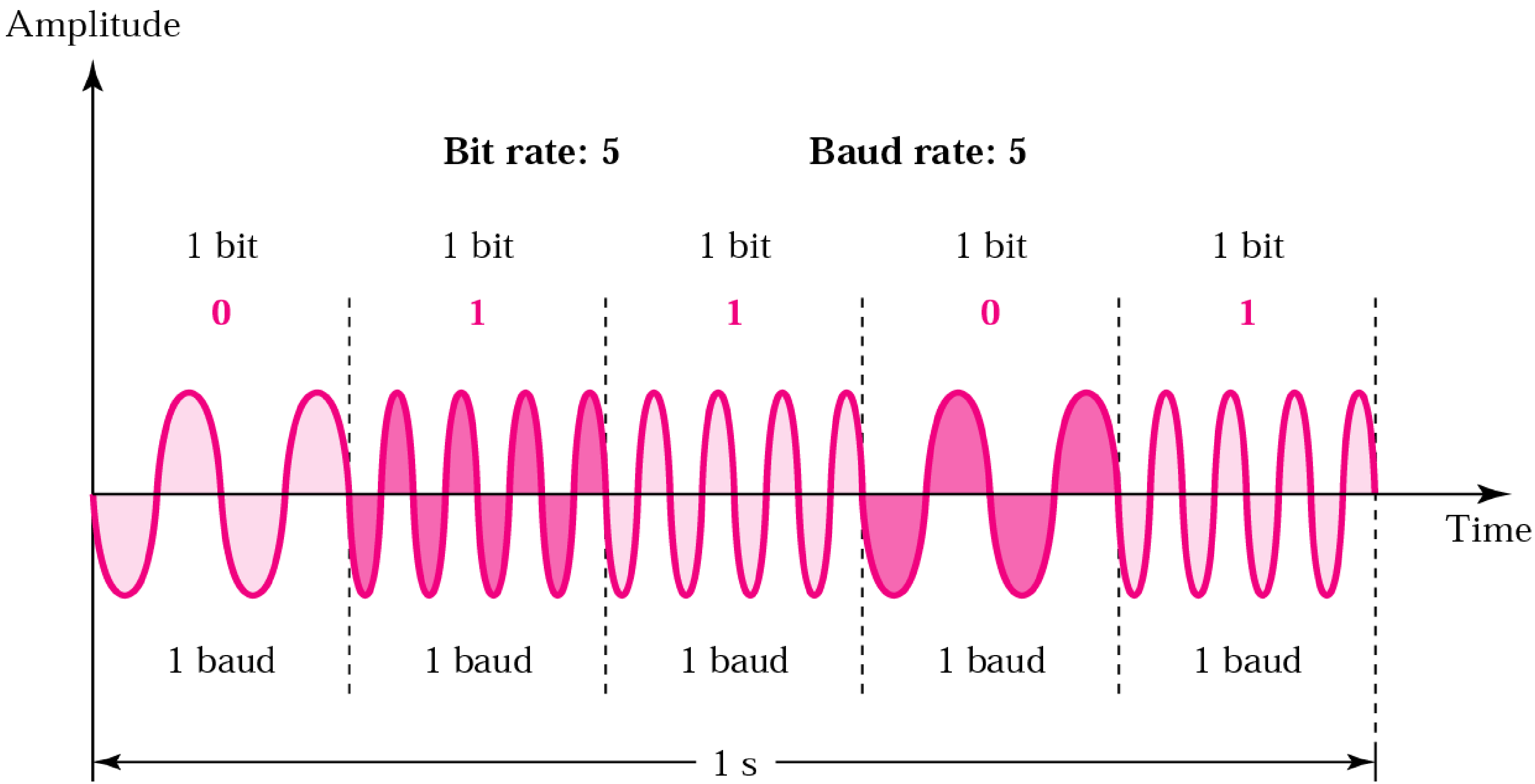
Peak amplitude during each bit duration is constant

ASK (contd.)

- Disadvantage
  - Highly susceptible to noise interference because ASK relies on amplitude to differentiate between 1 and 0
  - Need a great gap between amplitude values so that noise can be detected and removed
- OOK (on/off keying)
  - A popular ASK technique
  - Zero voltage represent a bit value (e.g., 0)
  - Save energy in transmitting information
- On voice-grade line, up to 1200bps
- Used to transmit digital data over optical fiber

Frequency Shift Keying (FSK)

Only frequency is varied to represent binary 1 or 0

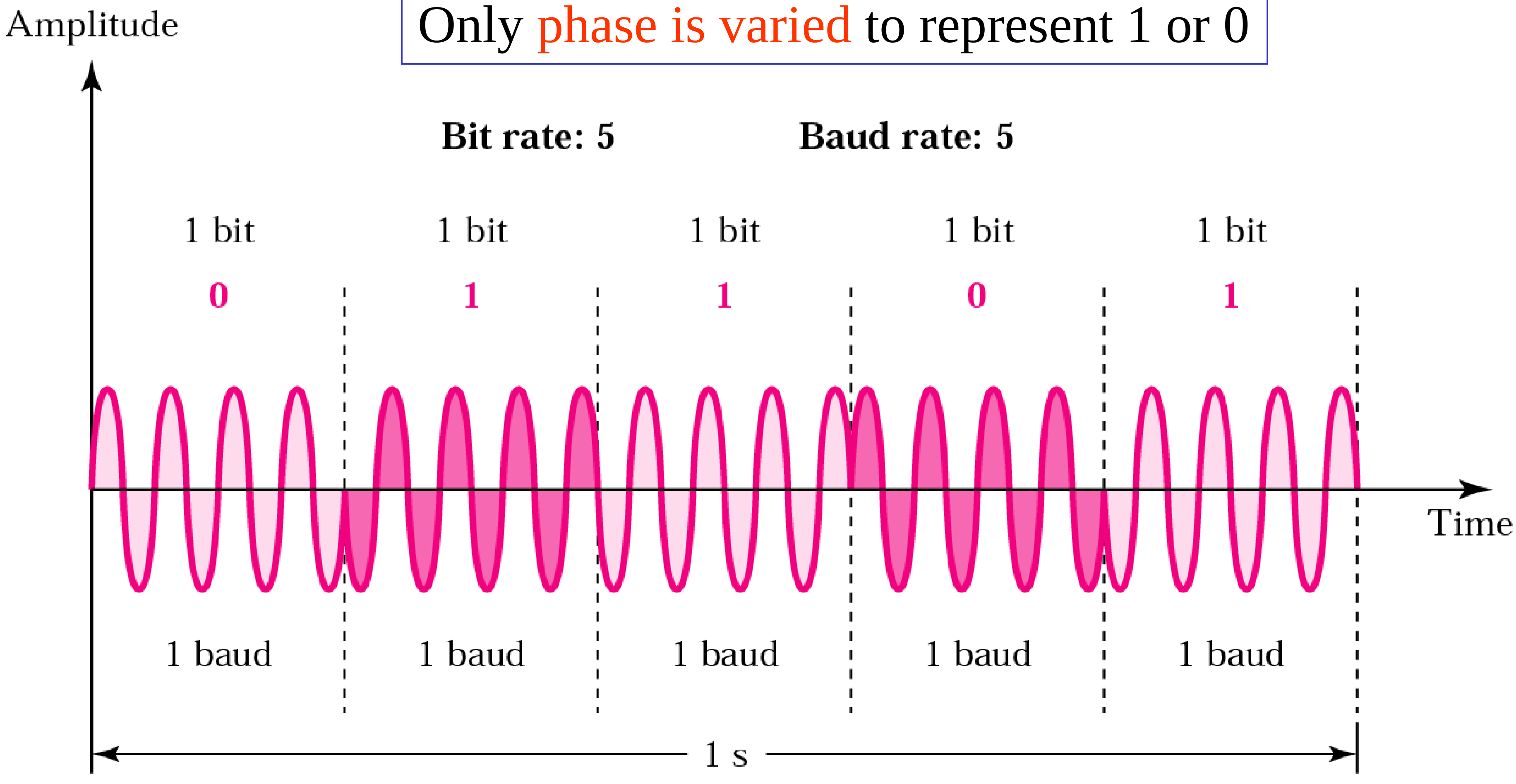


FSK vs. ASK

- FSK
  - Less susceptible to error
  - On voice-grade lines, up to 1200bps
  - Commonly used for high-freq (3-30 Mhz) radio
  - Also used at even high freq on LANs that use coaxial cable

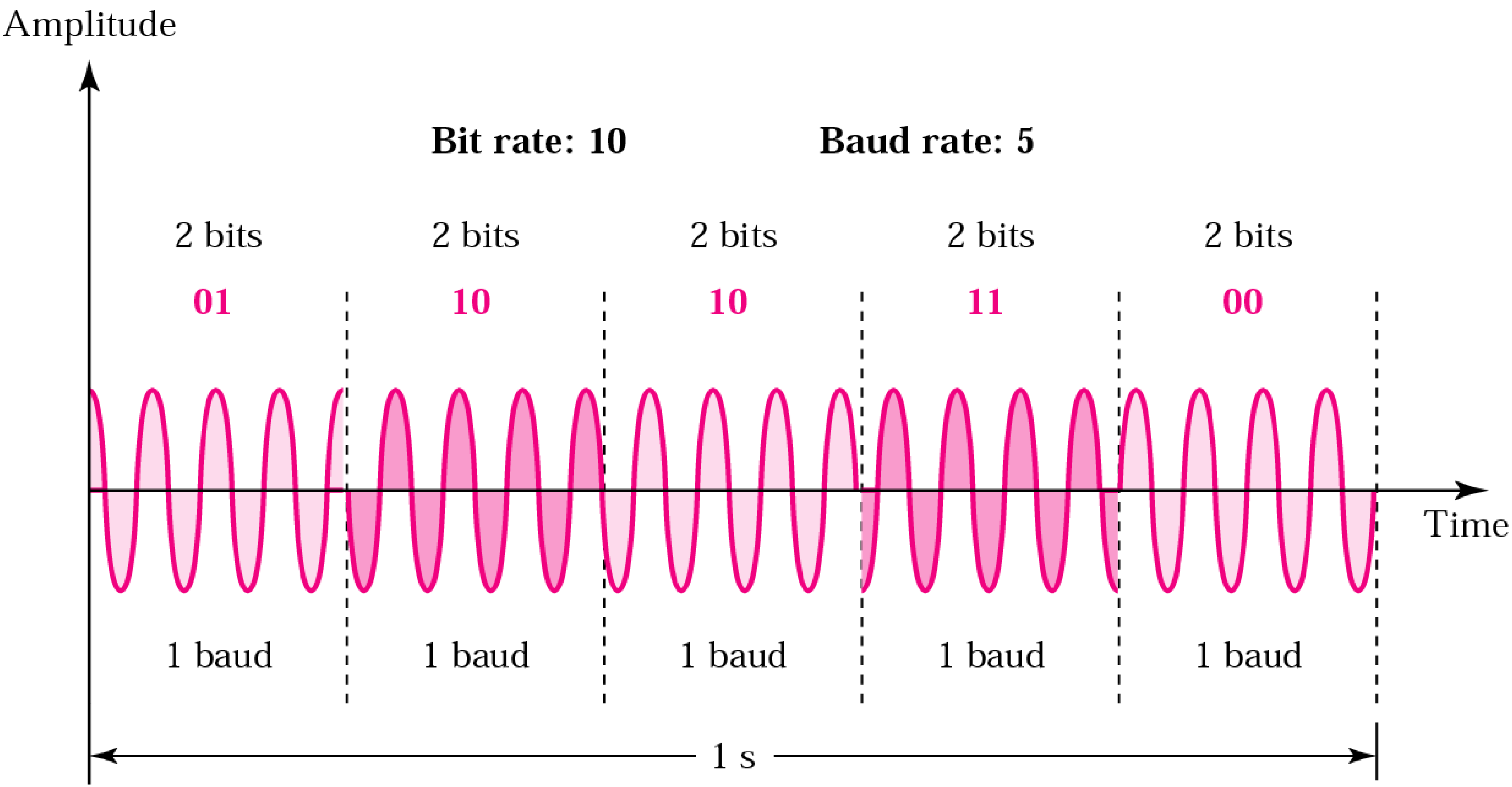
Phase Shift Keying (PSK)

Only phase is varied to represent 1 or 0



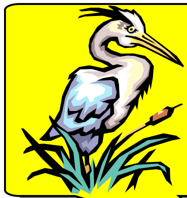
2-PSK: only 2 phase values are used, each for 1 or 0

4-PSK



Note: PSK is no susceptible to the noise degradation that affects ASK or to the bandwidth limitations of FSK

PSK: Drawback

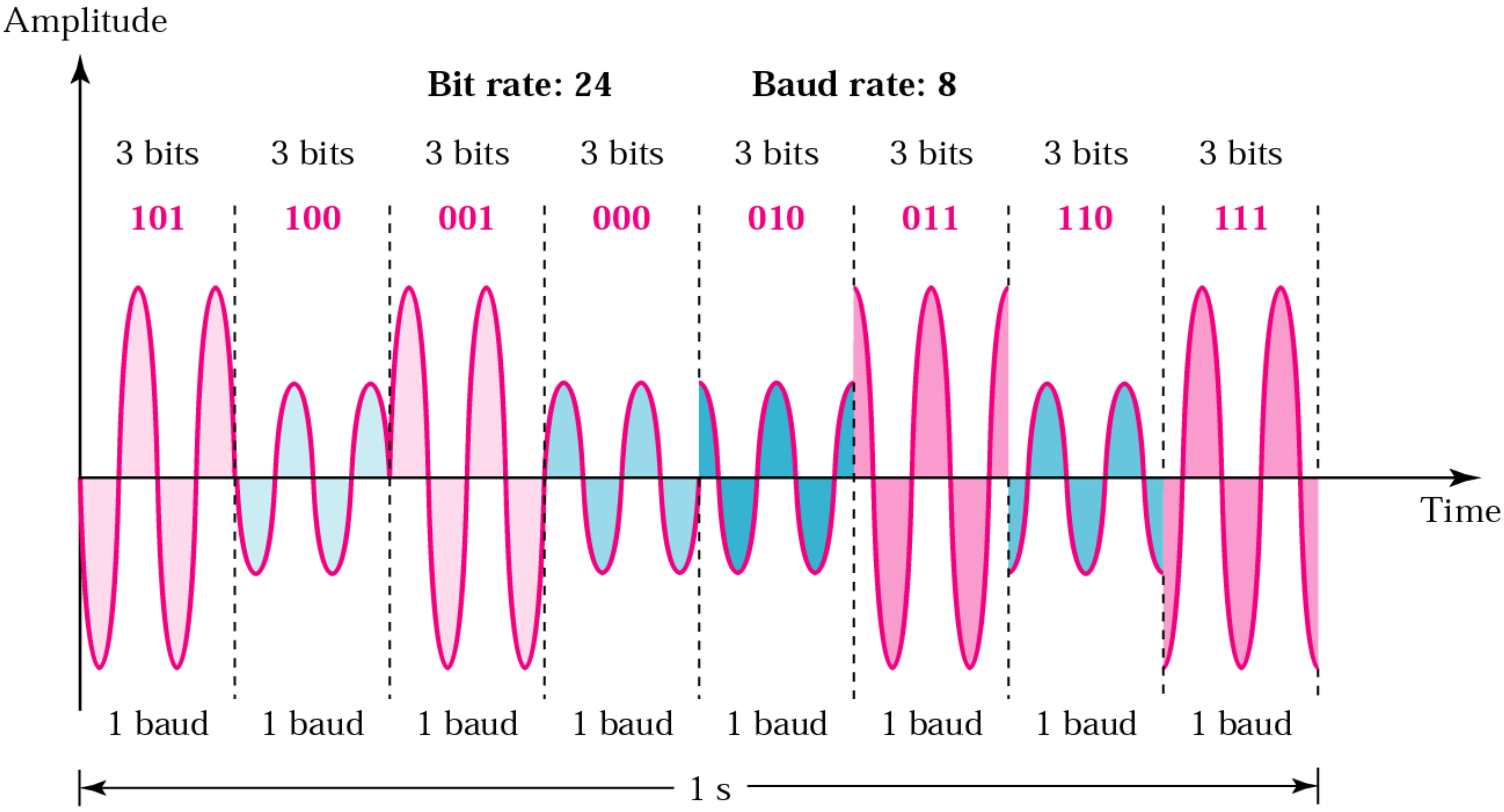


Note:

Modulation device is not able to distinguish small differences in phase => limit BitRate

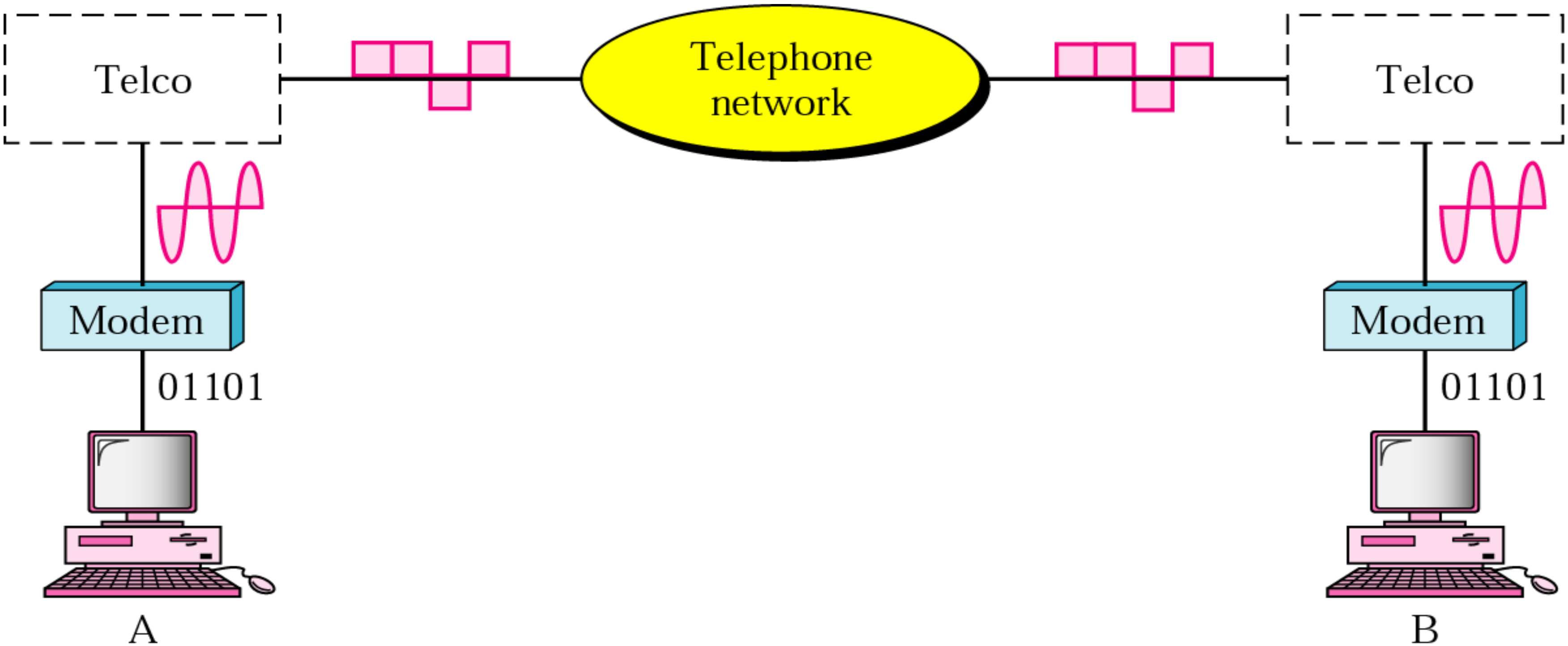
Why not combine PSK and ASK: **x** variations in phase with **y** variations in amplitude result in **xy** variations => increase bit rate

8-QAM

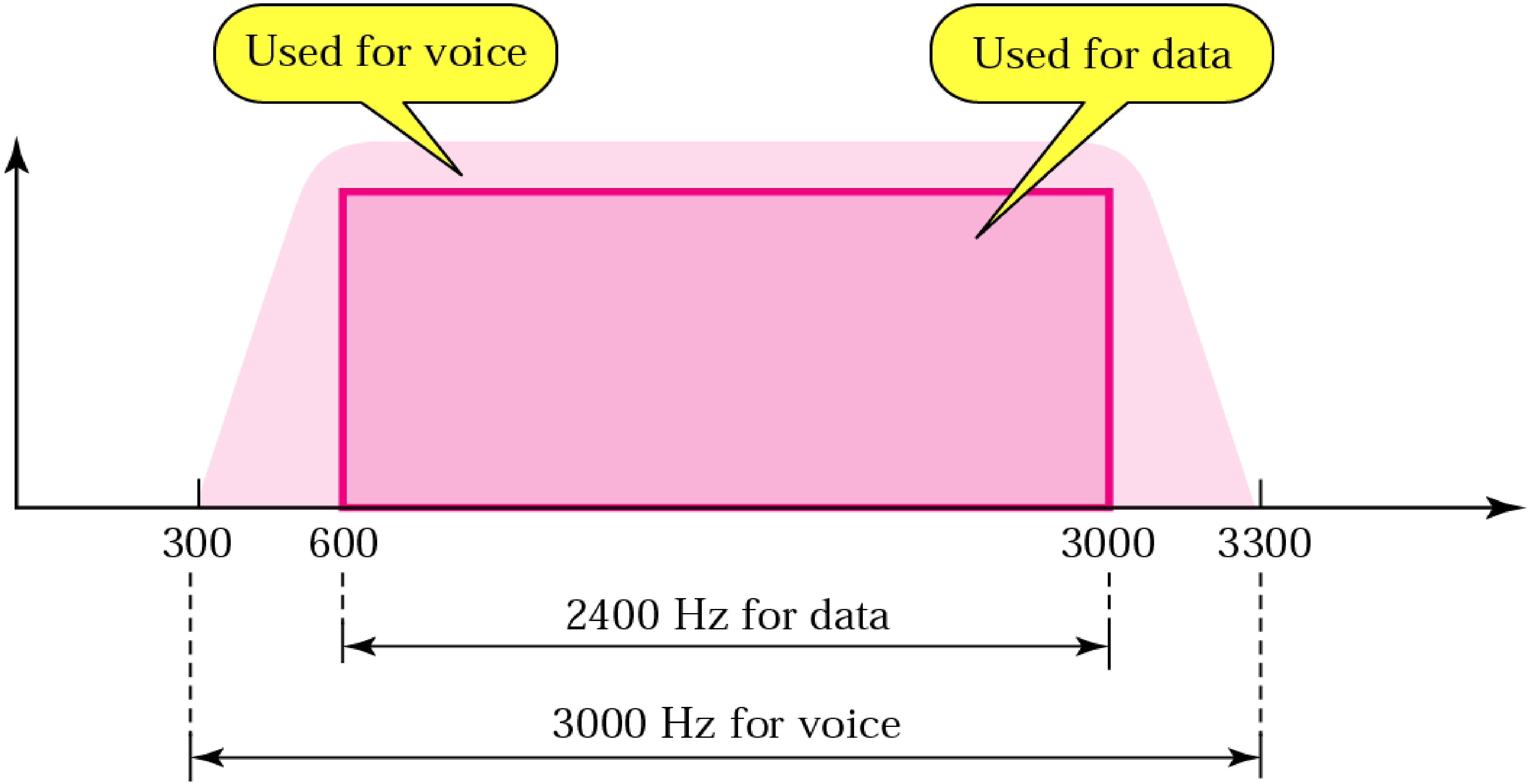

Amplitude
Bit rate: 24
Baud rate: 8
3 bits
3 bits
3 bits
3 bits
3 bits
3 bits
3 bits
3 bits
101
100
001
000
010
011
110
111
Time
1 baud
1 baud
1 baud
1 baud
1 baud
1 baud
1 baud
1 baud
1 s

Telephone Modem

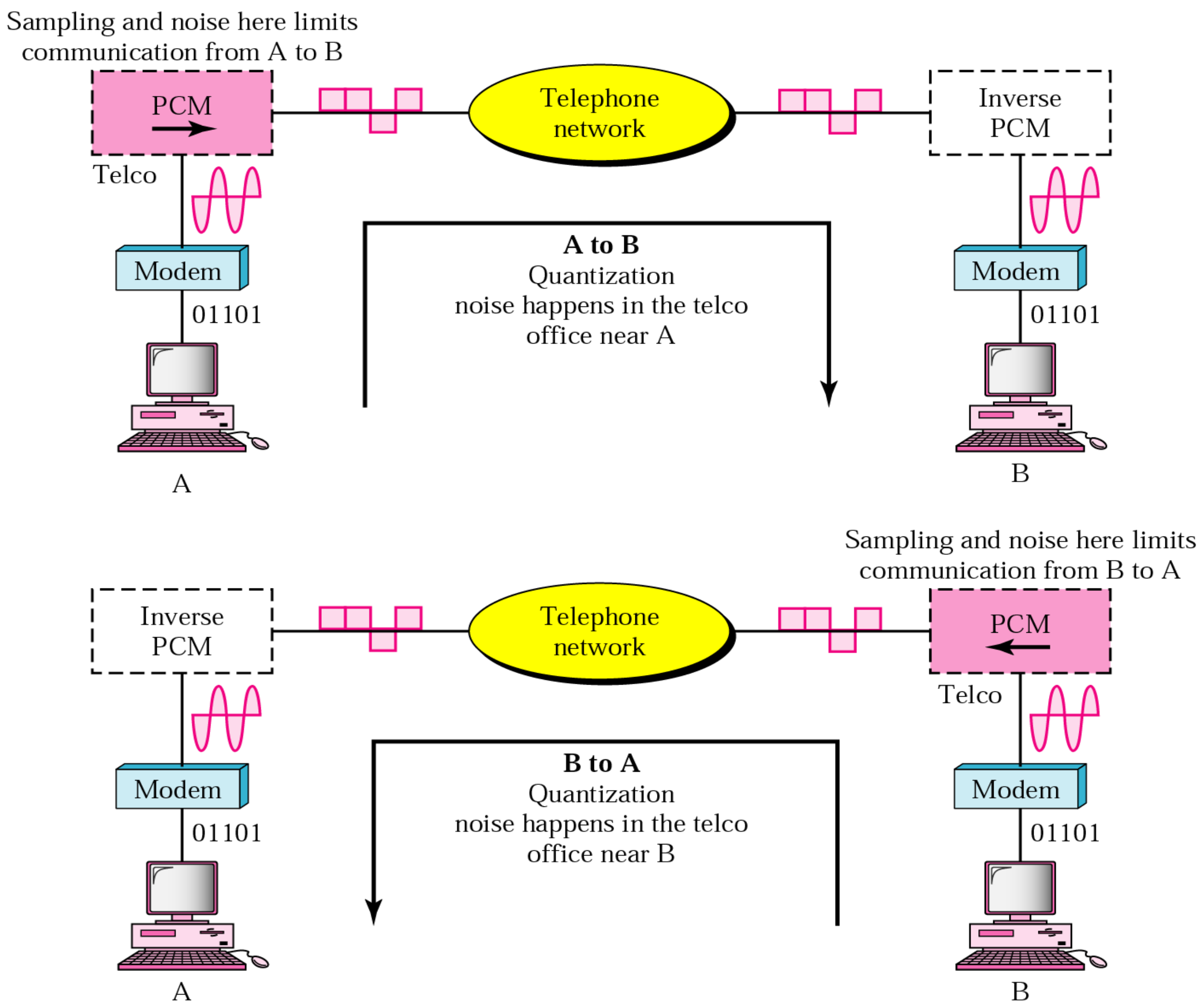
- A telephone line has a bandwidth of almost 2400 Hz for data transmission (600 – 3000 Hz). This bandwidth defines a baseband nature which needs to modulate for data transmission – modem.
- Modem – modulator and demodulator



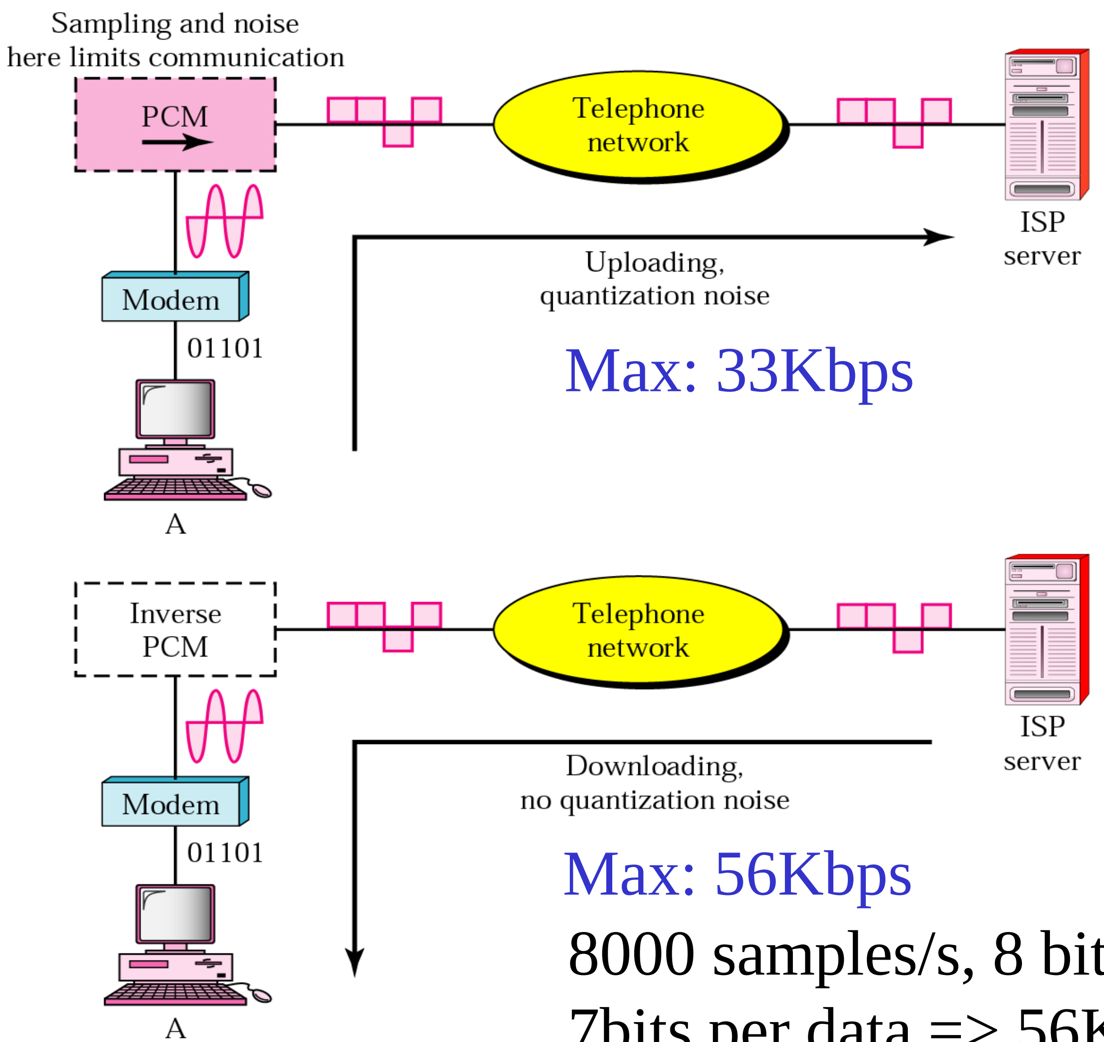
Telephone Line Bandwidth



Traditional Modems



56K Modem: V.90



8000 samples/s, 8 bits/sample,

7bits per data => 56Kbps

56K Modem: V.92

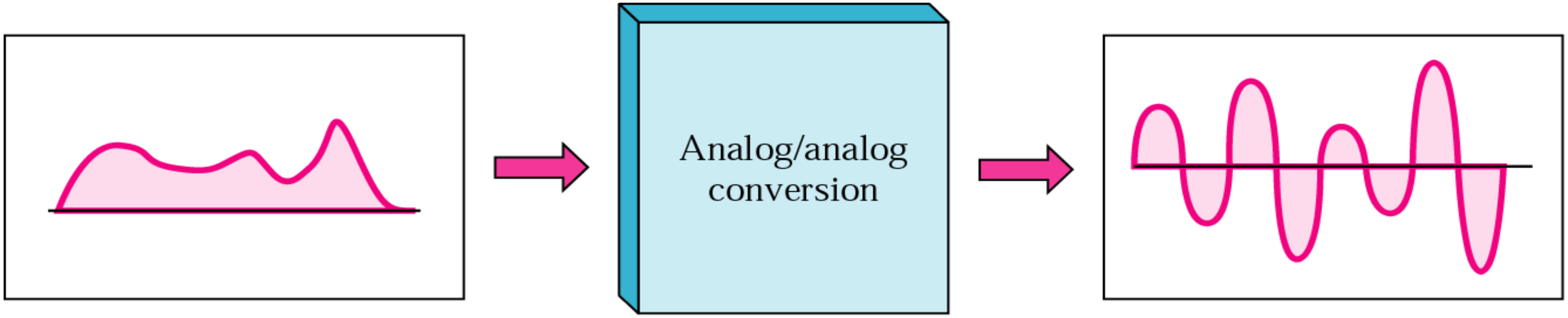
- Similar to V.90
  - Modem can adjust speed
  - If noise allows => upload max 48 Kbps, download still 56 Kbps
- V.92: can interrupt the Internet connection when there is an incoming call (if call-waiting service is installed)

Carrier Signal

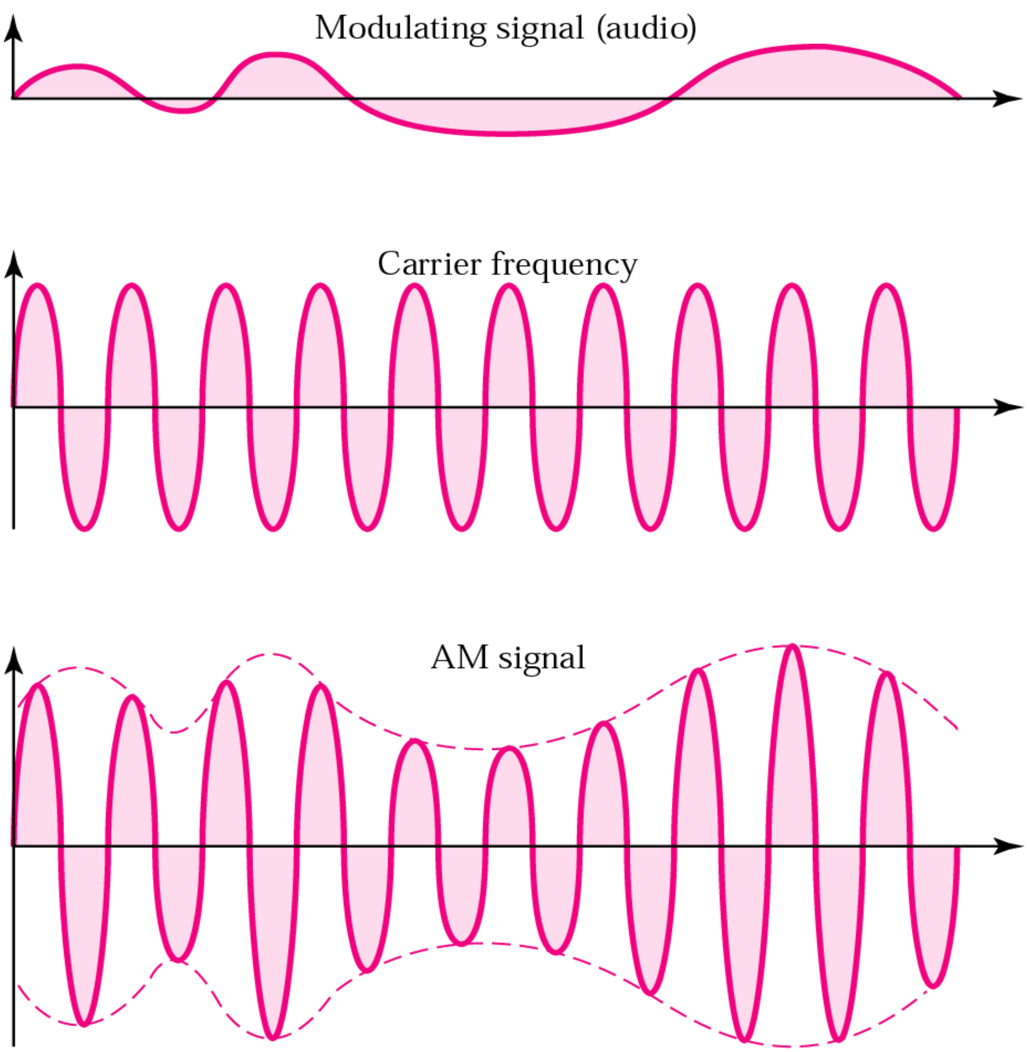
- The carrier signal is "the high frequency signal which carries the information (transmitting signal) through space as an electromagnetic wave".
- The carrier signal conveys the information.
- In general, carrier signals are high frequency sinusoidal pulses.
- These carrier signals allow the specific medium to transmit the input signal.

Analog to Analog Modulation

- Representation of analog information by an analog signal
- Why do we need it? Analog is already analog!!!
  - Because we may have to use a band-pass channel
  - Think about radio…
- Schemes
  - Amplitude modulation (AM)
  - Frequency modulation (FM)
  - Phase modulation (PM)



Amplitude Modulation: AM

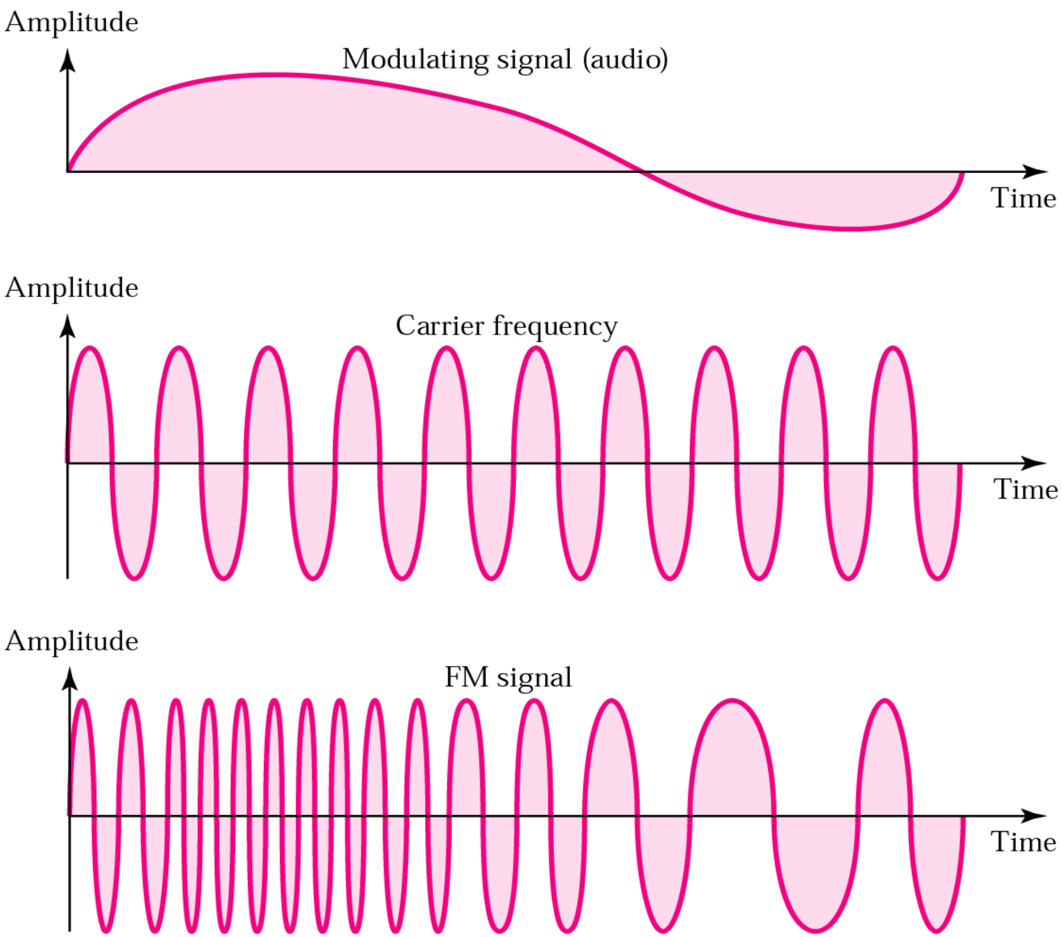


Carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal

Freq, phase remain same

$$s(t) = (1+n_a x(t))\cos(2\pi f_c t)$$

Frequency Modulation: FM



Freq. of carrier signal is modified to reflect the changing amplitudes of the modulating signal

Amp., phase remain same

Phase Modulation: PM

- Only phase is varied to reflect the change of amplitude in modulating signal
- Require simpler hardware than FM
  - Use in some systems as an alternative to FM

Phase Modulation: PM

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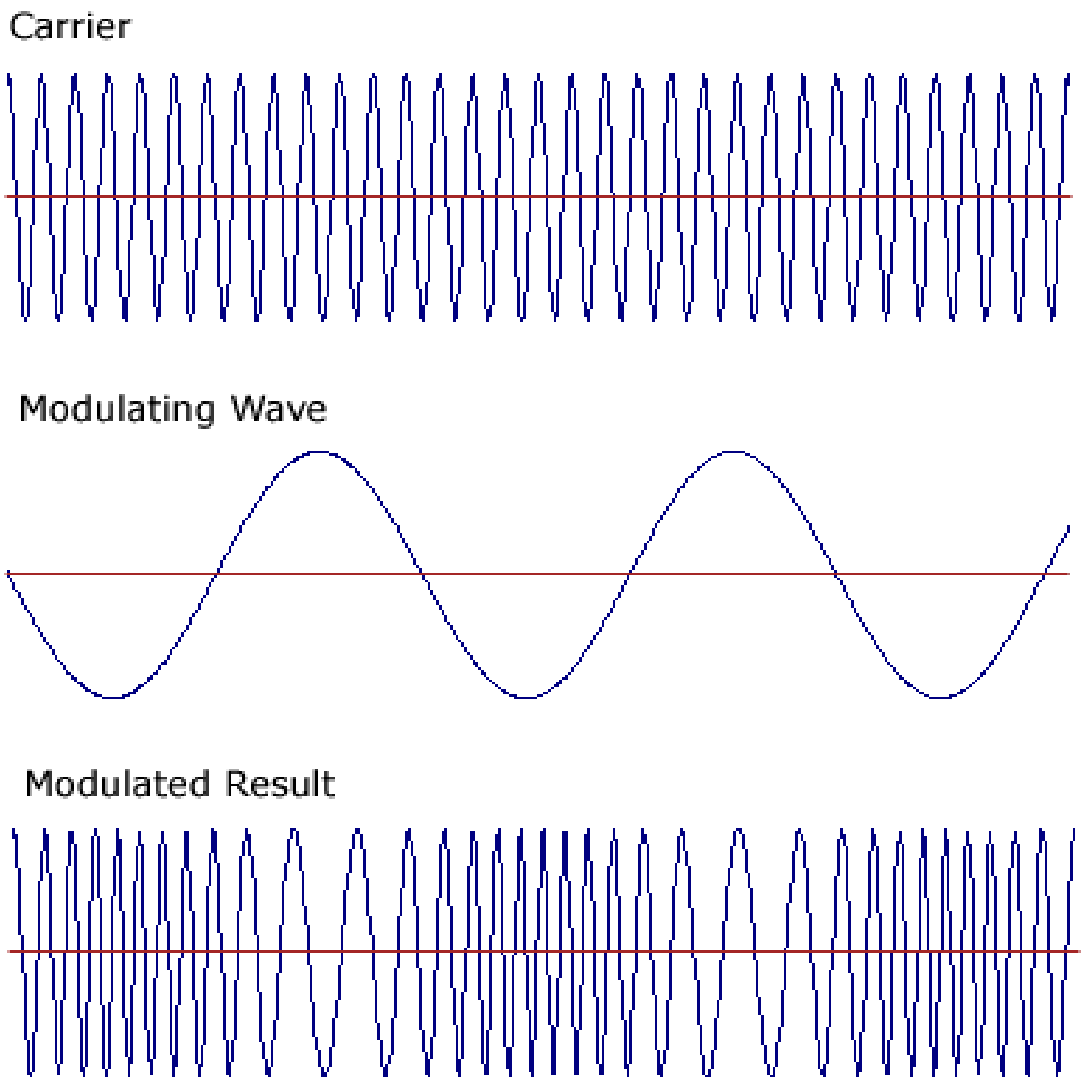

Thanks All